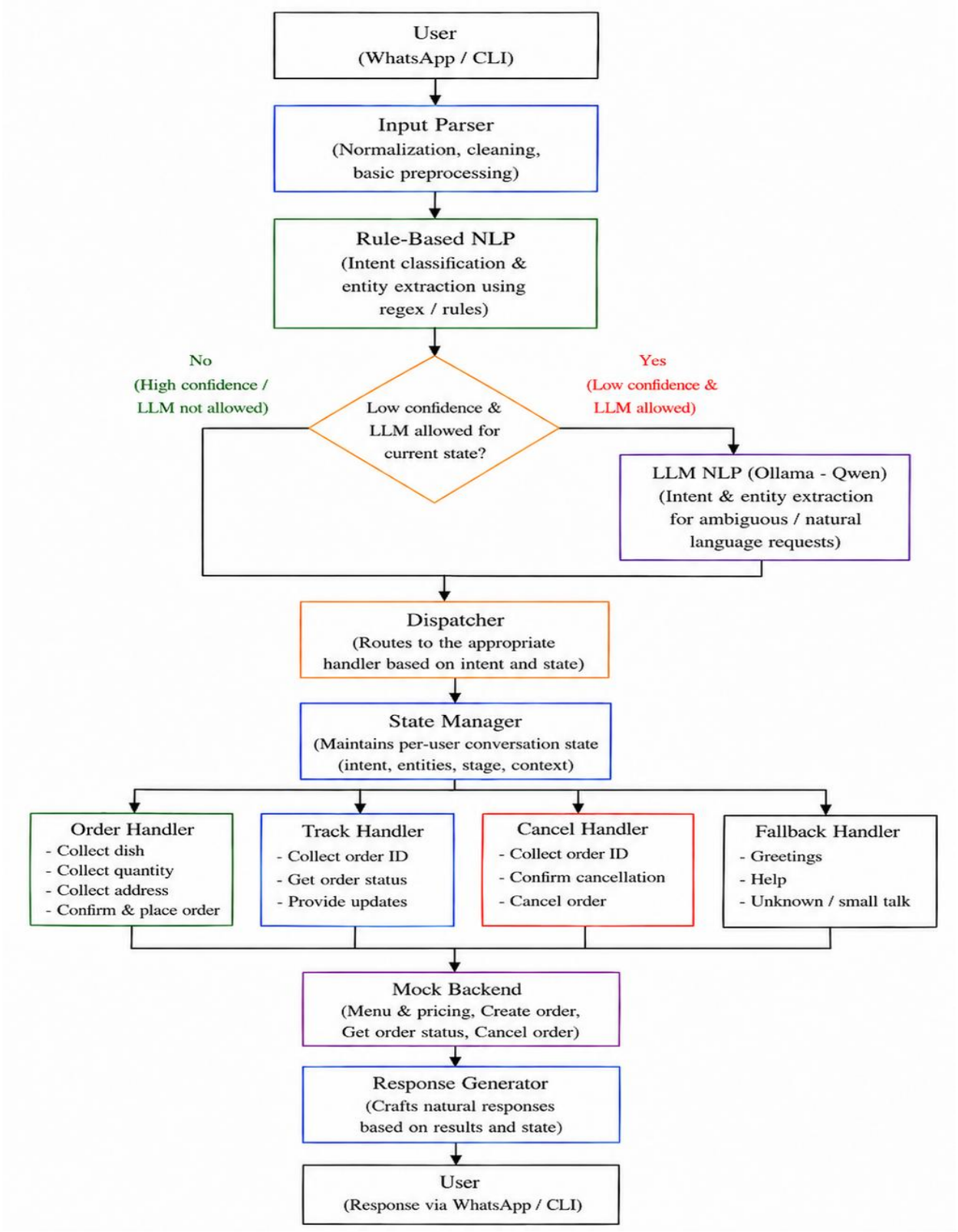
QuickBite Chatbot - Design Document

1. Overview

QuickBite is a food ordering chatbot that simulates a WhatsApp conversation. It allows users to place, track, and cancel orders through a multi turn chat. The system uses rule based NLP for clear requests and a local LLM to handle low confidence or more natural user messages

2. Architecture



3. Handling Ambiguity

QuickBite first checks the user message using rule-based NLP with regex, keywords, and menu matching. Based on the result, it assigns a confidence score. If the confidence is high, the result goes directly to the Dispatcher. If the confidence is low, the LLM can be used to better understand the request. For inputs such as quantity, address, order ID, and confirmation, the chatbot uses the current conversation state instead of the LLM.

4. Scaling to 100+ Menu Items

With 100+ menu items, managing regex patterns and aliases for every dish would be difficult. Instead, menu items can be stored as embeddings in a vector database. The user's request is also converted into an embedding and matched with the menu using similarity search. The top matching items are then sent to the LLM to select the best option or ask the user for clarification.

5. Potential Failures and Mitigation

Intent Misclassification. Rule-based classification may not always understand the different ways users express the same request. In the future, a model such as Logistic Regression, SVM, or Naive Bayes could be trained on past user messages, using TF-IDF to convert the text into features. Low-confidence results can still be sent to the LLM.